Terje Lafton Senior Platform Engineer

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Experience

Intility

Senior Platform Engineer
Aug 2022 - Present
Returned to the Intility network team, working mostly on Azure. First project was a PoC for Azure Virtual WAN to replace per-customer on-prem VPN tunnels, built first in Terraform and then converted to Crossplane after running into issues with CI/CD automation. The Virtual WAN was scrapped because of cost, but we kept using Crossplane.

Built a custom Crossplane Provider using gNMI and YANG to drive a new per-customer configuration across the core network from Kubernetes resources, with a frontend on top. Few teams use Kubernetes for network configuration, but the declarative model fits the problem better than typical imperative automation.

The 2.0 rebuild moved IPAM into NetBox, with a NetBox plugin as a low-level frontend, and replaced the Crossplane Provider with a custom Go Kubernetes Operator built around a shared reconciler pattern. The Operator uses OpenTelemetry from the start, with Logfire as the backend. A gRPC proxy in front of the routers caches gNMI operations, dropping router hits from thousands per second to near zero.

Deployed across nine clusters on Intility's new Kubernetes platform, with a management cluster running Argo and Kargo and lab/staging/preprod/prod separation. Lab and staging uses Cisco CML for device simulation; preprod runs read-only against prod routers.

Named in the Crossplane v1.16 release blog post for community contributions, mainly PR 5453, a proposal to change how resource statuses are handled in the Provider framework.

Oda

Software Developer
Jan 2022 - Jun 2022
Worked on the network and sites team during Oda's European expansion. The team handled both existing facilities and new locations. Replaced the Meraki setup with Cisco SDWAN, learning the platform and testing on routers in the facilities. Wrote Python to populate NetBox from operational systems, since documentation was minimal and scattered. The longer-term plan was to automate more of how new facilities get deployed.

Intility

Senior Engineer
Sep 2020 - Dec 2021
As I got more senior, work shifted from operational cases to project work and training newer engineers joining the network team. Owned the software side of Intility's Cisco ACI setup, building

Python tooling and APIs to automate tenant operations (creating APs, EPGs, and contracts) and fabric provisioning (new leaves and their port configurations). Python became a daily tool, with some Ansible work too.

Engineer
Mar 2016 - Aug 2020
Started in Intility's IT support function, where there was no first-line and everyone worked as second-line. Took the CCNA and moved into the Network Services NOC, working on Cisco core and customer networks across multiple data centres serving thousands of customers, including site-to-site VPNs between Intility's private cloud and customer Azure environments.

Norges Idrettsforbund

Apprentice/IT Support Technician
Aug 2013 - Feb 2016
Two-year apprenticeship: first-line phone support for in-house services used by 1M+ members, then on-site desk supporting sports federations and regional offices. Stayed on after the apprenticeship to lead the desk and train new apprentices.

Skills

Kubernetes: Custom Operators, Crossplane, Argo, Kargo, Openshift
Programming: Go, gRPC, Python, Scripting
Networking: gNMI, YANG, NetBox, Cisco (XR, CML, Meraki), Palo Alto, SDWAN
Cloud: Azure, Terraform
Observability: OpenTelemetry, Logfire

Certifications

Cisco Certified Network Associate (CCNA)
Aug 2017 - Jun 2020

Talks

From Terraform to Crossplane
Intility Developer Day, 2024
Reliability issues we ran into automating Terraform through CI/CD pipelines, with the broader point that automating a CLI tool isn't always a good idea. Then on to Crossplane: what we built with it, the value of a Kubernetes-style control plane, and the reconciler pattern you get for free.

Education

Hønefoss Videregående

IKT Servicefag Aug 2010 - Jun 2013